

Complete Formula Sheet — Matrices, Determinants, Linear Systems & Lines in the Plane

(Theory + Exercises, with explanations)

1. Matrices — Basic Notation

- **Matrix:** $A = (a_{ij})$, m rows $\times$ n columns. A table of numbers; a_{ij} is the entry at row i , column j — this notation lets any entry be addressed unambiguously.
- **Equality of matrices:** $A = B$ iff same dimension and $a_{ij} = b_{ij} \forall i,j$. Two tables represent the same information only if every corresponding cell matches.
- **Transpose A^T :** rows of A become columns of A^T . A "mirror" of the table across its main diagonal; useful for symmetric-matrix tests and the adjugate formula.
- $(A^T)^T = A$. Mirroring twice returns the original table.

2. Special Matrices

- **Row/column matrix:** $1 \times n$ or $n \times 1$. The simplest possible matrices, often representing a single vector.
- **Null matrix:** all entries 0. The "zero" of matrix addition.
- **Square matrix (order n):** $m = n$. Only square matrices can have an inverse or a single determinant number associated with the whole table.
- **Main diagonal:** entries a_{ii} . The diagonal line from top-left to bottom-right of a square matrix; many special matrix types are defined relative to it.
- **Symmetric matrix:** $A = A^T$. Reflecting the matrix across its main diagonal leaves it unchanged — off-diagonal entries mirror each other.
- **Diagonal matrix:** $a_{ij} = 0$ for $i \neq j$. All the "information" sits on the main diagonal; these matrices are easy to multiply and invert.
- **Identity matrix I_n :** diagonal with 1's on the diagonal. The matrix equivalent of the number 1 — multiplying by it changes nothing.
- **Upper/lower triangular matrix:** zeros below/above the diagonal. Common output of Gaussian elimination; their determinant is just the product of the diagonal entries.

3. Matrix Operations

- **Sum:** $(A+B)_{ij} = a_{ij} + b_{ij}$ (same dimension required). Combine two tables of the same shape by adding corresponding cells.
- **Scalar multiplication:** $(kA)_{ij} = k \cdot a_{ij}$. Scale every entry uniformly, like stretching every number by the same factor.
- **Matrix product:** $(A \cdot B)_{ij} = \sum_k a_{ik} b_{kj}$, requires $\text{columns}(A) = \text{rows}(B)$. Each output entry is a "dot product" of a row of A with a column of B — this is why the inner dimensions must match.

- **Product is not commutative:** $A \cdot B \neq B \cdot A$ in general. Order matters because the roles of "rows" and "columns" being combined are not symmetric.
- $A \cdot I_n = I_n \cdot A = A$. The identity matrix acts like the number 1 under multiplication.
- **Inverse matrix:** $A \cdot A^{-1} = A^{-1} \cdot A = I_n$. The matrix "undoes" A's transformation, the way $1/x$ undoes multiplication by x .
- **Singular matrix:** a matrix with no inverse. Analogous to trying to divide by zero — some matrices simply have no "undo."

4. Determinants — Direct Computation

- **Order 1:** $|a_{11}| = a_{11}$. A 1×1 matrix's determinant is just the number itself.
- **Order 2:** $|A| = a_{11}a_{22} - a_{12}a_{21}$. The product of the main diagonal minus the product of the anti-diagonal.
- **Order 3 (Sarrus):** $|A| = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{23}a_{12} - a_{13}a_{22}a_{31} - a_{23}a_{32}a_{11} - a_{33}a_{21}a_{12}$. A memorable diagonal pattern (extend the matrix with copies of its first two columns and multiply along the diagonals) that only works for 3×3 matrices.

5. Determinants — Cofactor Expansion (order ≥ 4)

- **Adjugate (cofactor) A_{ij} :** $(-1)^{i+j}$ times the minor obtained by deleting row i and column j . Captures "how much" the rest of the matrix determines the determinant once one row and column are set aside, with a sign that alternates in a checkerboard pattern.
- **Cofactor expansion along row i :** $|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}$. Breaks a large determinant into a weighted sum of smaller ones — choosing a row/column with many zeros minimizes the work.
- **Cofactor expansion along column j :** $|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \dots + a_{nj}A_{nj}$. The same idea applied by column instead of row — the determinant is the same either way.

6. Properties of Determinants

- $\det(A) = \det(A^T)$. Transposing doesn't change how "spread out" or "collapsed" the matrix's transformation is.
- **Swapping two rows/columns flips the sign.** A determinant tracks orientation as well as size; swapping two directions reverses orientation.
- **Equal/proportional rows or columns $\Rightarrow \det = 0$.** If two rows carry the same direction, the matrix "collapses" a dimension, giving zero volume/area.
- **A zero row/column $\Rightarrow \det = 0$.** A row of all zeros means one direction contributes nothing, collapsing the volume to zero.
- **Scaling one row/column by k scales the whole determinant by k .** Stretching one direction by k stretches the resulting volume/area by exactly k .
- **A row/column written as a sum splits the determinant into two.** Determinants are linear in each row/column separately, so a sum inside one row can be pulled out as a sum of two determinants.

- **A row/column that's a linear combination of others $\Rightarrow \det = 0$.** Same underlying reason as the equal/proportional case — no genuinely new direction is added.
- **Adding a multiple of one row/column to another doesn't change det.** This "shearing" operation doesn't change the enclosed volume/area, which is exactly why it's safe to use during Gaussian-elimination-style simplification.
- **$\det(A \cdot B) = \det(A) \cdot \det(B)$.** Applying one transformation then another scales volumes by the product of their individual scaling factors.

7. Inverse Matrix via the Adjugate

- **Existence condition:** A has an inverse $\Leftrightarrow \det(A) \neq 0$. A zero determinant means the matrix collapses space into a lower dimension, an operation with no unique "undo."
- **Formula:** $A^{-1} = \text{adj}(A)^T / \det(A)$, where $\text{adj}(A)$ replaces every entry with its adjugate. Built entirely from cofactors and the determinant, this closed-form expression avoids needing to "guess" the inverse.

8. Rank of a Matrix

- **Minor of order h :** determinant from selecting h rows and h columns. A smaller "sample" square block taken from within a possibly non-square matrix.
- **Bordered minor:** a minor of order $h+1$ obtained by adding one aligned row and column to an existing minor. Grows a known non-zero minor step-by-step to search for a larger one.
- **Rank theorem:** $\text{rk}(A)$ = order of the largest non-zero minor. The rank measures how many truly independent directions/equations the matrix encodes — found by growing minors until none larger survive as non-zero.
- **Vectors independence test:** k vectors in $\mathbb{R}^n$ (as rows/columns of A) are linearly independent $\Leftrightarrow \text{rk}(A) = k$; in particular, n vectors in $\mathbb{R}^n$ are independent $\Leftrightarrow \det(A) \neq 0$. Independence means no vector can be built from the others — exactly what a full-rank (or non-zero determinant) matrix guarantees.

9. Systems of Linear Equations — Notation

- **General form:** $a_{i1}x_1 + \dots + a_{in}x_n = b_i$ for $i=1, \dots, m$. A collection of straight-line (linear) conditions the unknowns must simultaneously satisfy.
- **Matrix form:** $A \cdot X = B$. Packs coefficients (A), unknowns (X) and constants (B) into a single matrix equation, turning system-solving into matrix algebra.
- **Augmented matrix:** $M = (A|B)$. Keeps every piece of information about the system in one table for the purposes of rank comparison.
- **Equivalent systems:** share the same solution set. Different-looking systems can secretly describe the same geometric/algebraic answer.

10. Classifying Systems — Rouché–Capelli Theorem

- $\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{M}) = n \Rightarrow$ **independent** (unique solution). *Enough independent conditions to pin down every unknown exactly.*
- $\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{M}) = r < n \Rightarrow$ **dependent**, with $(n-r)$ degrees of freedom. *Fewer independent conditions than unknowns leaves some unknowns free to vary.*
- $\text{rk}(\mathbf{A}) < \text{rk}(\mathbf{M}) \Rightarrow$ **inconsistent**. *The constant terms introduce a condition the coefficients alone cannot satisfy — a contradiction.*

11. Homogeneous Systems

- $\mathbf{A}\mathbf{X} = \mathbf{0}$ is **always consistent** (trivial solution $\mathbf{X} = \mathbf{0}$). *Plugging in all zeros always balances an equation whose right-hand side is also zero.*
- **Non-trivial solutions exist** $\Leftrightarrow \text{rk}(\mathbf{A}) < n$. *Only when there's "room" (a degree of freedom) can a solution other than all-zeros satisfy every equation.*

12. Gaussian Elimination

- **Row operations:** swap rows, scale a row by a non-zero constant, add a multiple of one row to another, delete an all-zero row. *These operations change the matrix's appearance without changing the underlying system's solution set.*
- **Row-echelon form:** a "staircase" pattern of leading entries. *Once in this shape, the system can be solved top-to-bottom trivially.*
- **Back-substitution:** solve the last equation, substitute upward, repeat. *Each step removes one unknown, using previously found values to simplify the next equation.*
- $\text{rk}(\mathbf{M}) =$ **number of non-zero rows in echelon form**. *Row operations don't change rank, so counting surviving non-zero rows reveals it directly.*

13. Cramer's Rule

- **Cramer's system:** square (n equations, n unknowns) with $\det(\mathbf{A}) \neq 0$. *Guarantees, by the existence condition for $\mathbf{A}^{-1}$, that a unique solution exists.*
- **Formula:** $x_i = \det(\mathbf{A}_i) / \det(\mathbf{A})$, where $\mathbf{A}_i$ is $\mathbf{A}$ with column i replaced by $\mathbf{B}$. *Each unknown is isolated by comparing how much replacing one column with the constants "changes" the determinant relative to the original.*
- **Dependent-system adaptation:** pick a non-zero $r \times r$ minor, treat variables outside it as parameters, and apply Cramer's rule to the reduced $r \times r$ system for each fixed parameter value. *Turns an underdetermined system into a genuine Cramer's system by temporarily "freezing" the extra unknowns.*

14. Equations of a Line in the Plane

- **Director vector:** $\mathbf{v} = \mathbf{Q} - \mathbf{P} = (v_1, v_2)$, from two points $\mathbf{P}, \mathbf{Q}$ on the line. *Encodes the line's direction — any vector parallel to the line works equally well.*
- **Vectorial equation:** $\mathbf{X} = \mathbf{P} + k\mathbf{v}$, $k \in \mathbb{R}$. *Every point on the line is the starting point $\mathbf{P}$ shifted by some multiple of the direction $\mathbf{v}$.*

- **Parametric equations:** $x = p_1 + k \cdot v_1$, $y = p_2 + k \cdot v_2$. Splits the vectorial equation into its two coordinate components.
- **Continuous equation:** $(x - p_1)/v_1 = (y - p_2)/v_2$ ($v_1, v_2 \neq 0$). Eliminates the parameter k by solving for it in both coordinate equations and setting the results equal.
- **Slope:** $m = v_2/v_1$. The ratio of vertical to horizontal change along the direction vector — how steep the line is.
- **Slope-intercept equation:** $y - p_2 = m(x - p_1)$. A line through a known point with a known slope.
- **Explicit equation:** $y = mx + n$, $n = -m \cdot p_1 + p_2$. The most common form, showing directly the slope m and the y -intercept n (where the line crosses the y -axis).
- **General equation:** $Ax + By + C = 0$, with $A = v_2$, $B = -v_1$, $C = -v_2 p_1 + v_1 p_2$. A single linear equation with no fractions or special cases (works even for vertical lines, where the slope is undefined).

15. Geometric Interpretation of SLE

- **2 variables → lines in the plane;** discussion of m lines via Rouché–Capelli. Each linear equation in x, y traces a straight line, so solving the system means finding common points among lines.
- **3 variables → planes in space;** discussion of m planes via Rouché–Capelli. Extends the same logic one dimension up: each equation now traces a flat plane.
- **Independent ($rk=n$):** unique intersection point. Exactly enough conditions to pin down a single point.
- **Dependent, 1 d.o.f. ($rk=2, n=3$):** planes meet along a common line. One missing condition leaves a whole line of solutions.
- **Dependent, 2 d.o.f. ($rk=1, n=3$):** all planes coincide. Two missing conditions leave an entire plane of solutions.
- **Inconsistent:** no common point (parallel/non-coincident planes, or planes with no shared intersection). The equations contradict each other geometrically — the surfaces never all meet.
- **A line in space as an intersection:** representable as a dependent 2-equation, 3-variable system (two non-coincident planes meeting along it). A line has "one degree of freedom," matching exactly a system with one fewer independent equation than unknowns.

16. Extra Techniques for Exercises

- **Solving $X \cdot A = B$:** $X = B \cdot A^{-1}$ (right-multiply by A^{-1}). Isolates X the way dividing both sides of a scalar equation isolates a variable — but the side of multiplication matters here.
- **Matrix polynomials** (e.g., $A^2 - 2A$): compute $A \cdot A$ first, then subtract $2A$ entrywise. Combines the product and scalar-multiplication rules in sequence.

- **Parametric determinant = 0:** solve the resulting polynomial equation in the parameter and factor it. *Turns a single numeric check into an equation whose roots mark the special/boundary cases.*
 - **Leontief production model:** $X = C \cdot X + D \Rightarrow X = (I - C)^{-1}D$. *Combines basic algebraic rearrangement with the inverse-matrix formula to solve for production levels given demand.*
 - **Testing if a point lies on a line:** substitute its known coordinate(s) into the parametric or continuous equations and solve for k; use that k for any missing coordinate(s). *Turns the abstract equation of a line into a concrete numeric check for one specific point.*
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Master Quick-Reference Table

#	Formula	Topic
1	$A = (a_{ij}), \text{ dimension } m \times n$	Matrices
2	$(A^T)^T = A$	Matrices
3	$A+B, kA, A \cdot B$ (with dimension rules)	Matrix operations
4	$A \cdot A^{-1} = A^{-1} \cdot A = I_n$	Inverse matrix
5	Order 2: $a_{11}a_{22} - a_{12}a_{21}$	Determinants
6	Order 3: Sarrus' rule	Determinants
7	$A_{ij} = (-1)^{i+j} \cdot (\text{minor})$	Cofactor/adjugate
8	$ A = \sum_j a_{ij} A_{ij}$ (row/col expansion)	Determinants
9	$\det(A) = \det(A^T)$; row swap flips sign; k-scaling; sum-splitting; linear combo $\Rightarrow 0$; row ops preserve det; $\det(AB) = \det(A)\det(B)$	Determinant properties
10	$A^{-1} = \text{adj}(A)^T / \det(A)$	Inverse via adjugate
11	$\text{rk}(A) = \text{order of largest non-zero minor}$	Rank
12	k vectors independent $\Leftrightarrow \text{rk}(\text{matrix}) = k$	Linear independence
13	$A \cdot X = B; M = (A B)$	SLE notation
14	$\text{rk}(A) = \text{rk}(M) = n \Rightarrow \text{independent}; r < n \Rightarrow \text{dependent } (n-r \text{ dof}); \text{rk}(A) < \text{rk}(M) \Rightarrow \text{inconsistent}$	Rouché-Capelli
15	$A \cdot X = 0$ always consistent; non-trivial sols $\Leftrightarrow \text{rk}(A) < n$	Homogeneous systems
16	Row-echelon form + back-substitution	Gaussian elimination
17	$x_i = \det(A_i) / \det(A)$	Cramer's rule
18	$X = P + k \cdot v$ (vectorial)	Line in the plane
19	$x = p_1 + kv_1, y = p_2 + kv_2$	Parametric line
20	$(x - p_1)/v_1 = (y - p_2)/v_2$	Continuous line
21	$y - p_2 = m(x - p_1), m = v_2/v_1$	Slope-intercept line

#	Formula	Topic
22	$y=mx+n$	Explicit line
23	$Ax+By+C=0$	General line equation
24	Independent→point; dependent(1 dof)→line; dependent(2 dof)→coincident planes; inconsistent→no common point	Geometric interpretation
25	$X \cdot A = B \implies X = B \cdot A^{-1}$	Matrix equation solving
26	$X = (I - C)^{-1}D$	Leontief model